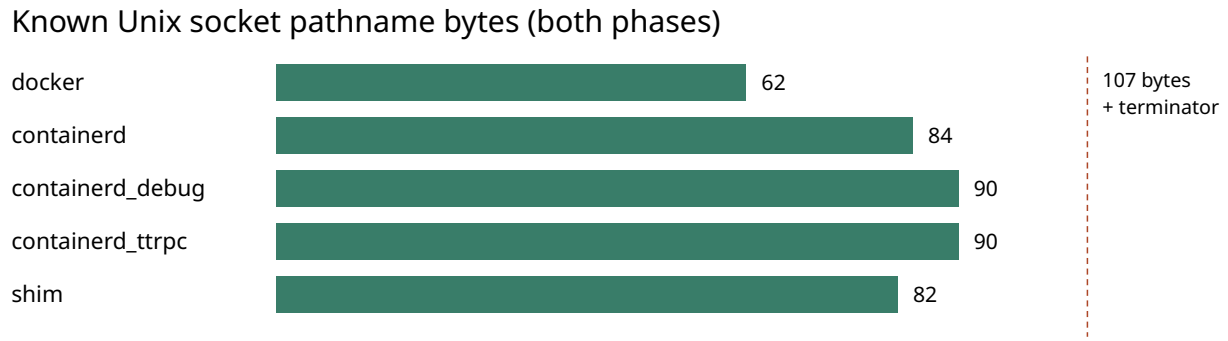
Private runtime supervisor v2: path and custody checks

The implementation shortens private runtime paths and separates assembly and qualification runtime directories on the retained disk. The five known Unix socket paths require 62, 84, 90, 90 and 82 pathname bytes in each phase; with a terminator, each fits the Linux 108-byte bound. The old Docker, managed containerd and debug paths required 113, 133 and 139 bytes.



Source-derived paths; 29 producer controls and 29 independent controls passed. No runtime execution.

Containerd 2.3.5 uses /run/containerd/s for root-owned shim sockets. The supervisor now masks that host path with a namespace-local read-only 4 KiB tmpfs, then binds an owned directory from the existing 4 GiB ext4 disk over it inside the service. It checks filesystem type, read-only mask, device and inode identity, and restores the mask before unmounting the disk. An uncertain bind or unbind outcome is retained and blocks continuation. Separate phase directory names prevent reusing assembly runtime directories during qualification.

Producer validation and the independent reviewer each passed 29 software controls. External command effects were mocked; no systemd service, Docker daemon, mount or container was executed. The reviewer also contributed design advice, so this is an exact source review and independent replay, not a blind review. The existing separate 600-second assembly and 300-second qualification limits remain unchanged; both admission drafts remain false.

Intermediate results are retained: inherited suite 20 passed and 1 error; expanded suite 27 passed and 1 error, both caused by the old predecessor-path fixture; corrected fixture 28 passed; phase-edit suite 23 passed and 6 errors from an argument-count regression; final corrected suite 29 passed. The review also caught fixed-directory reuse before final freezing. Source snapshots and raw records identify each revision.

The inventory covers Docker API, managed containerd gRPC/debug/ttrpc and hashed shim sockets. Optional metrics, libnetwork plugin and runc console paths are not exhaustively inventoried. Installed software was identified as Docker 25.0.16 and containerd 2.3.5; retained primary source references are separately hashed. Live namespace, cgroup, filesystem and daemon enforcement still needs the separate admitted runtime qualification.